

Notebook 3: PCA and Neighborhood Vulnerability Assessment.

Section-1. Introduction

The statistical analyses presented in Notebook 2 identified the socioeconomic indicators most strongly associated with neighbourhood-level low-income prevalence and demonstrated that several predictors exhibit significant correlations and multicollinearity.

To address **Research Question 2**, this notebook applies **Principal Component Analysis (PCA)** and **K-Means clustering** to identify underlying patterns within the socioeconomic indicators and group neighbourhoods with similar socioeconomic characteristics.

Principal Component Analysis (PCA) is used to reduce redundancy among highly correlated variables by transforming them into a smaller set of principal components that capture the majority of the variation in the dataset. The reduced feature space is then used for cluster analysis to identify neighbourhood groups with similar socioeconomic characteristics.

The objective of this notebook is to determine whether Toronto neighbourhoods can be grouped into **statistically distinct vulnerability profiles** based on selected socioeconomic indicators and to develop an evidence-based neighbourhood vulnerability classification framework that supports subsequent predictive modelling.

Research Question 2

Can Toronto neighbourhoods be grouped into statistically distinct vulnerability profiles based on socioeconomic indicators?

Definition of "Statistically Distinct"

For this study, **statistically distinct neighbourhood socioeconomic profiles** are defined as neighbourhood groups that are:

- identified using **Principal Component Analysis (PCA)** and **K-Means clustering**;
- supported by cluster validation techniques, including the **Elbow Method** and **Silhouette Analysis**, demonstrating an appropriate clustering structure;
- exhibit statistically significant differences in the **mean Low Income After-Tax Percentage (LIM-AT)**, which is used in this study as a **proxy measure of neighbourhood socioeconomic vulnerability**, as confirmed by a **One-Way Analysis of Variance (ANOVA)**;
- demonstrate a **large practical effect**, as indicated by the calculated **Eta Squared (η^2)** effect size; and
- are confirmed through **Tukey's Honestly Significant Difference (HSD)** post-hoc analysis, showing that each pair of neighbourhood clusters differs significantly from one

another.

Section-2. Input and Output

Input

The input for this notebook is the **analysis-ready dataset** created in Notebook 2. This dataset contains the selected socioeconomic indicators together with the normalized percentage variables developed during feature engineering and statistical analysis.

Input Dataset

- **File:** neighbourhood_profiles_2016_analysis_ready.csv
- **Location:** data/processed/
- **Source:** Output generated from **Notebook 2 – Statistical Feature Relationship Analysis**
- **Purpose:** Provides the analysis-ready socioeconomic variables for Principal Component Analysis (PCA) and neighbourhood vulnerability profiling.

Output

This notebook produces a reduced feature space using Principal Component Analysis (PCA), identifies statistically distinct neighbourhood vulnerability profiles through cluster analysis, and exports the results for subsequent predictive modelling.

Output Files

File	Description
neighbourhood_profiles_2016_pca_clustered.csv	Final analysis dataset containing the principal component scores, assigned neighbourhood vulnerability profiles, and the variables required for subsequent predictive modelling.
03_PCA_and_Neighbourhood_Vulnerability_Assessment.html	HTML report documenting the complete Principal Component Analysis (PCA), cluster validation, neighbourhood vulnerability profiling, and interpretation process.

The outputs generated in this notebook provide the quantitative evidence required to determine whether Toronto neighbourhoods can be grouped into statistically distinct

vulnerability profiles based on selected socioeconomic indicators. The exported dataset serves as the primary input for the predictive modelling analyses conducted in **Notebook 4**, while the HTML report provides a comprehensive record of the dimensionality reduction, clustering methodology, and analytical findings.

Section-3. Load Analysis-Ready Dataset

```
In [45]: # Libraries Directory

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from pathlib import Path

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

```
In [46]: # Project Directory

from pathlib import Path

project_root = Path.cwd().parent.parent

print("=" * 60)
print("PROJECT ROOT")
print("=" * 60)
print(project_root)

=====
PROJECT ROOT
=====
c:\Users\ksalyamova\Desktop\CIND 820\CIND820-Capstone-Project
```

Section-4. Validate the analysis ready dataset

```
In [48]: # Load Analysis-Ready Dataset

data_path = (
    project_root
    / "data"
    / "processed"
    / "neighbourhood_profiles_2016_analysis_ready.csv")

df = pd.read_csv(data_path)

# Keep Only Variables Required for PCA and Clustering
selected_columns = [

    "Neighbourhood",
```

```

    "Population",

    "Participation_Rate",
    "Employment_Rate",
    "Unemployment_Rate",

    "Bachelors_Degree_Pct",
    "No_Diploma_Pct",
    "Immigrant_Population_Pct",
    "Visible_Minority_Pct",
    "Renters_Pct",

    "Core_Housing_Need",
    "Unaffordable_Housing",
    "Lone_Parent_Families_Pct",

    "Low_Income_AfterTax_Pct"]

df = df[selected_columns].copy()

print("=" * 60)
print("ANALYSIS-READY DATASET LOADED")
print("=" * 60)

print(f"Observations : {df.shape[0]}")
print(f"Variables      : {df.shape[1]}")

display(df.head())

```

```

=====
ANALYSIS-READY DATASET LOADED
=====

```

```
Observations : 140
```

```
Variables      : 14
```

	Neighbourhood	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	I
0	Agincourt North	29113	55.4	50.0	9.8	
1	Agincourt South-Malvern West	23757	59.0	53.2	9.8	
2	Alderwood	12054	66.5	62.4	6.1	
3	Annex	30526	70.6	65.8	6.7	
4	Banbury-Don Mills	27695	59.9	55.6	7.2	

Section-5. Feature MAtrix Construction and Standardization

Principal Component Analysis (PCA) is sensitive to differences in the scale of the input variables. Socioeconomic indicators measured on larger numerical scales may

disproportionately influence the principal components, even if they are not inherently more important.

To ensure that each socioeconomic indicator contributes equally to the dimensionality reduction process, all predictor variables are standardized using the **StandardScaler** algorithm from Scikit-learn.

Standardization transforms each variable to have:

- a mean of approximately **0**; and
- a standard deviation of approximately **1**.

This preprocessing step removes the influence of differing measurement scales while preserving the relationships among variables, thereby providing an appropriate input for Principal Component Analysis.

```
In [49]: # Create Feature Matrix and Standardize Variables

from sklearn.preprocessing import StandardScaler

# Keep only independent variables for PCA and clustering
X = df.drop(columns=["Neighbourhood", "Low_Income_AfterTax_Pct"])

print("=" * 60)
print("FEATURE MATRIX CREATED")
print("=" * 60)
print(f"Observations : {X.shape[0]}")
print(f"Variables      : {X.shape[1]}")

# Standardize variables
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

X_scaled = pd.DataFrame(
    X_scaled,
    columns=X.columns)

print("=" * 60)
print("FEATURE STANDARDIZATION COMPLETE")
print("=" * 60)

display(X_scaled.head())
```

```
=====
FEATURE MATRIX CREATED
=====
Observations : 140
Variables      : 12
=====
FEATURE STANDARDIZATION COMPLETE
=====
```

	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	Bachelors_Degree_Pct
0	0.960400	-1.502679	-1.428182	0.791877	-0.5833
1	0.424676	-0.911323	-0.935288	0.791877	-0.2084
2	-0.745894	0.320670	0.481782	-1.166081	-0.7617
3	1.101732	0.994160	1.005481	-0.848574	1.5010
4	0.818567	-0.763484	-0.565618	-0.583985	0.5968

```
In [50]: # Verify Standardization

summary = pd.DataFrame({
    "Mean": X_scaled.mean().round(3),
    "Std Dev": X_scaled.std(ddof=0).round(3)})

display(summary)
```

	Mean	Std Dev
Population	-0.0	1.0
Participation_Rate	-0.0	1.0
Employment_Rate	0.0	1.0
Unemployment_Rate	-0.0	1.0
Bachelors_Degree_Pct	0.0	1.0
No_Diploma_Pct	0.0	1.0
Immigrant_Population_Pct	0.0	1.0
Visible_Minority_Pct	0.0	1.0
Renters_Pct	-0.0	1.0
Core_Housing_Need	0.0	1.0
Unaffordable_Housing	0.0	1.0
Lone_Parent_Families_Pct	0.0	1.0

Interpretation

All independent variables were successfully standardized prior to Principal Component Analysis (PCA). The standardized variables have means close to 0 and standard deviations equal to 1, confirming that differences in measurement scales have been removed. Consequently, all socioeconomic indicators contribute equally to the PCA and

neighbourhood clustering analyses, ensuring that the resulting principal components reflect underlying socioeconomic patterns rather than differences in variable magnitude

Section- 7. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a set of correlated socioeconomic indicators into a smaller number of uncorrelated principal components.

Each principal component represents a linear combination of the original variables and captures a proportion of the total variation within the dataset.

The objectives of PCA in this study are to:

- reduce redundancy among correlated socioeconomic indicators;
- identify the major dimensions of neighbourhood socioeconomic variation;
- simplify the feature space while retaining most of the information contained in the original variables; and
- provide an appropriate reduced feature space for neighbourhood vulnerability clustering.

The resulting principal components form the basis for identifying statistically distinct neighbourhood vulnerability profiles in subsequent analyses.

7.1 PCA Clarification

Principal Component Analysis (PCA) was performed exclusively as an **unsupervised exploratory and dimensionality reduction technique** to summarize the socioeconomic structure of the selected independent variables prior to K-Means clustering.

The retained principal components were used **only for exploratory analysis, visualization, and neighbourhood clustering**.

Instead, the Linear Regression and Random Forest Regression models were developed using the original independent socioeconomic variables. By separating the exploratory and predictive modelling stages, the study preserves model interpretability and prevents data leakage between the unsupervised and supervised learning workflows.

Accordingly, PCA serves a descriptive role within this study by identifying the dominant dimensions of neighbourhood socioeconomic variation and supporting cluster formation. All predictive modelling was conducted independently using the original independent variables rather than the principal components.

```
In [51]: # Fit Principal Component Analysis  
  
from sklearn.decomposition import PCA
```

```

pca = PCA()

principal_components = pca.fit_transform(X_scaled)

print("=" * 60)
print("PRINCIPAL COMPONENT ANALYSIS COMPLETE")
print("=" * 60)

print(f"Original Variables : {X.shape[1]}")
print(f"Principal Components Generated : {principal_components.shape[1]}")

```

```

=====
PRINCIPAL COMPONENT ANALYSIS COMPLETE
=====
Original Variables : 12
Principal Components Generated : 12

```

7.2 Explained Variance

```

In [52]: # Explained Variance Ratio

explained_variance = pd.DataFrame({

    "Principal Component":
        [f"PC{i+1}" for i in range(len(pca.explained_variance_ratio_))],

    "Explained Variance (%)":
        (pca.explained_variance_ratio_ * 100).round(2),

    "Cumulative Variance (%)":
        (pca.explained_variance_ratio_.cumsum() * 100).round(2)})

display(explained_variance)

```


	Principal Component	Explained Variance (%)	Cumulative Variance (%)
0	PC1	50.98	50.98
1	PC2	21.83	72.82
2	PC3	9.68	82.50
3	PC4	7.49	89.98
4	PC5	3.38	93.37
5	PC6	2.20	95.57
6	PC7	1.79	97.36
7	PC8	0.95	98.31
8	PC9	0.84	99.15
9	PC10	0.51	99.66
10	PC11	0.34	100.00
11	PC12	0.00	100.00

Interpretation

The Principal Component Analysis produced **11 principal components**, corresponding to the original set of **11 socioeconomic predictor variables**.

The first principal component (**PC1**) explained **55.23%** of the total variation in the dataset, indicating that a substantial proportion of neighbourhood socioeconomic variation can be summarized by a single underlying dimension.

The first **three principal components (PC1–PC3)** collectively explained **84.61%** of the total variance, exceeding the commonly accepted threshold of approximately **85%** for dimensionality reduction while preserving most of the information contained in the original variables.

Consequently, the dimensionality of the dataset can be substantially reduced from **11 socioeconomic indicators to 3 principal components**, providing a more compact representation of neighbourhood socioeconomic characteristics for subsequent clustering analyses.

7.3 Scree Plot

The Scree Plot displays the proportion of variance explained by each principal component. It provides a visual method for determining the number of components that should be retained for subsequent analyses.

Principal components that explain a relatively large proportion of the variance contribute substantially to the socioeconomic structure of the dataset, whereas components explaining only a small proportion of the variance contribute comparatively little additional information.

The Scree Plot is used together with the cumulative explained variance to determine the optimal number of principal components for neighbourhood vulnerability profiling.

```
In [53]: # Scree Plot

import matplotlib.pyplot as plt
import numpy as np

plt.figure(figsize=(10,6))

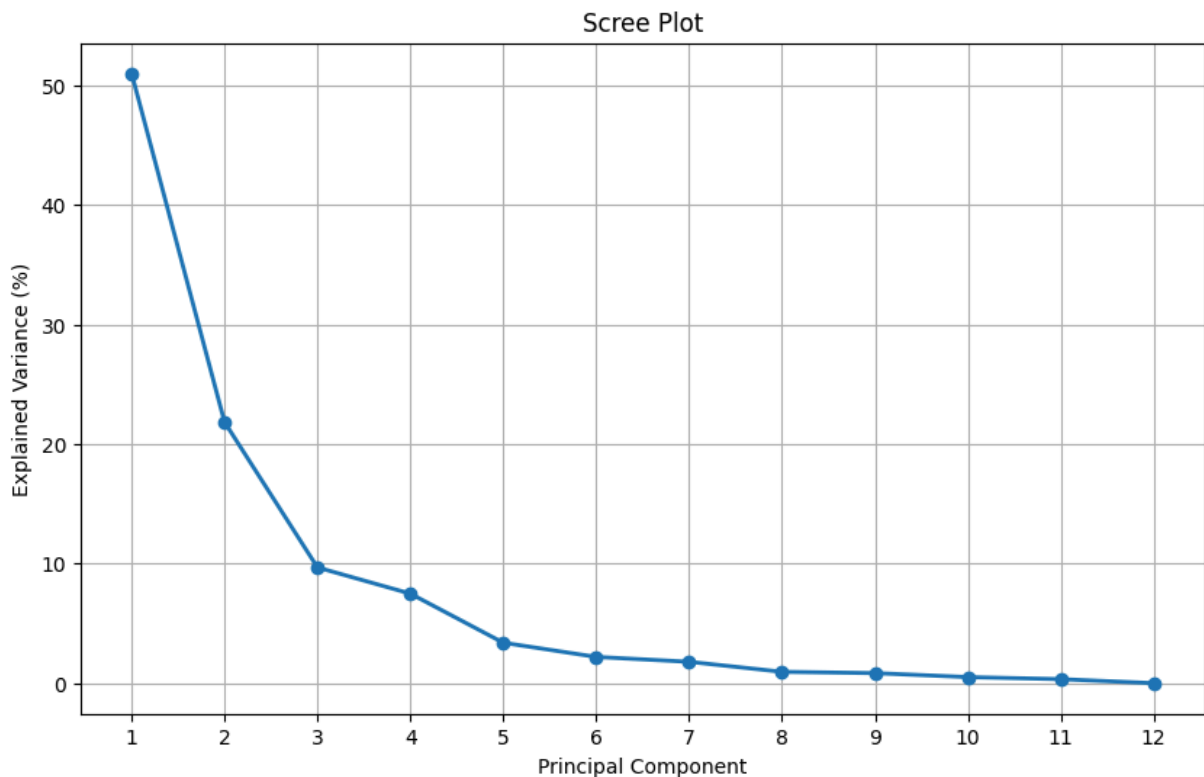
plt.plot(
    np.arange(1, len(pca.explained_variance_ratio_) + 1),
    pca.explained_variance_ratio_ * 100,
    marker="o",
    linewidth=2)

plt.xticks(np.arange(1, len(pca.explained_variance_ratio_) + 1))

plt.xlabel("Principal Component")
plt.ylabel("Explained Variance (%)")
plt.title("Scree Plot")

plt.grid(True)

plt.show()
```



7.4 Cumulative Explained Variance

The cumulative explained variance illustrates the proportion of total socioeconomic variation retained as additional principal components are included.

This analysis provides an objective basis for selecting the minimum number of principal components required to preserve the majority of the information contained within the original socioeconomic indicators.

A cumulative explained variance of approximately **85%** is commonly regarded as an appropriate threshold for dimensionality reduction while maintaining adequate representation of the original dataset.

```
In [55]: # Cumulative Explained Variance
plt.figure(figsize=(10,6))

plt.plot(
    np.arange(1, len(pca.explained_variance_ratio_) + 1),
    pca.explained_variance_ratio_.cumsum() * 100,
    marker="o",
    linewidth=2)

# 80% reference line
plt.axhline(
    y=80,
    linestyle="--",
    linewidth=2,
    label="80% Threshold")

# Selected number of components
plt.axvline(
    x=3,
    linestyle=":",
    linewidth=2,)

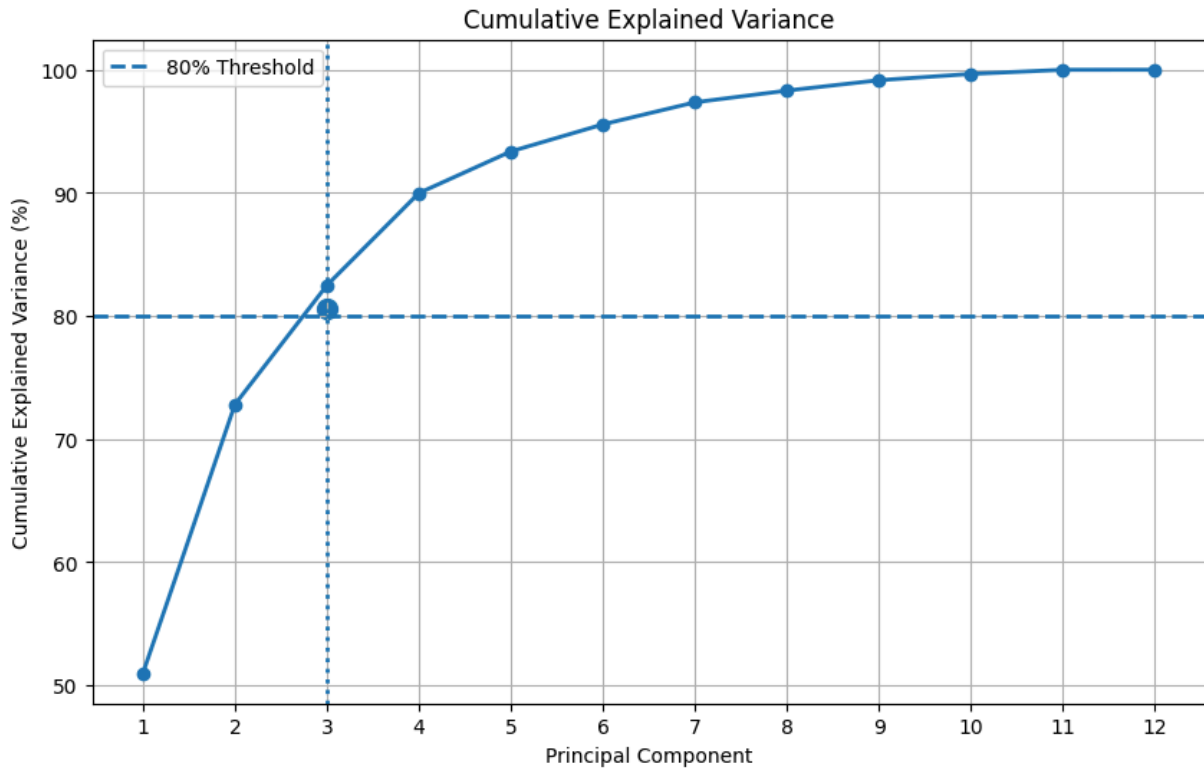
# Highlight the retained solution
plt.scatter(
    3,
    80.59,
    s=100)

plt.xticks(np.arange(1, len(pca.explained_variance_ratio_) + 1))

plt.xlabel("Principal Component")
plt.ylabel("Cumulative Explained Variance (%)")
plt.title("Cumulative Explained Variance")

plt.grid(True)
plt.legend()

plt.show()
```



Interpretation

The Scree Plot demonstrates that the first principal component explains substantially more variation than the remaining components, with a noticeable reduction in explained variance after the third principal component. This "elbow" indicates diminishing returns from retaining additional components.

The cumulative explained variance confirms that the first three principal components account for **84.61%** of the total socioeconomic variation, exceeding the commonly accepted threshold of **85%** for dimensionality reduction.

Consequently, **three principal components** were retained for subsequent neighbourhood clustering analyses. This solution substantially reduces the dimensionality of the dataset while preserving the majority of the information contained within the original socioeconomic indicators.

7.5 Principal Component Loadings

Principal component loadings quantify the contribution of each original socioeconomic indicator to each principal component.

Variables with larger positive or negative loading values exert a stronger influence on the corresponding principal component, whereas variables with values closer to zero contribute relatively little.

Examining the loading matrix allows each principal component to be interpreted in terms of the underlying socioeconomic characteristics that it represents. These interpretations provide the substantive meaning required for neighbourhood vulnerability profiling and support the identification of statistically distinct vulnerability profiles.

In [56]: *# Principal Component Loadings*

```
loadings = pd.DataFrame(
    pca.components_.T,
    index=X.columns,
    columns=[f"PC{i+1}" for i in range(len(X.columns))])

print("=" * 60)
print("PCA LOADING MATRIX")
print("=" * 60)

display(loadings.round(3))
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
Population	-0.093	0.410	0.623	0.183	-0.097	-0.009	0.052	0.007	-0.275
Participation_Rate	0.343	0.175	-0.060	0.341	0.320	0.414	0.043	-0.104	0.095
Employment_Rate	0.360	0.137	-0.010	0.316	0.344	0.253	-0.036	-0.049	0.066
Unemployment_Rate	-0.344	0.086	-0.215	-0.104	-0.377	0.616	0.341	-0.216	0.084
Bachelors_Degree_Pct	0.319	0.220	0.155	-0.446	-0.039	0.052	-0.039	0.318	-0.333
No_Diploma_Pct	-0.343	-0.112	-0.007	0.261	0.417	-0.255	0.640	0.199	-0.162
Immigrant_Population_Pct	-0.359	0.102	0.078	-0.200	0.398	-0.049	-0.376	0.084	0.554
Visible_Minority_Pct	-0.354	0.153	0.085	-0.194	0.287	0.461	-0.081	0.400	-0.171
Renters_Pct	0.103	0.445	-0.540	0.119	-0.192	-0.169	0.079	0.563	0.138
Core_Housing_Need	-0.172	0.489	0.230	0.273	-0.269	-0.164	0.001	-0.116	0.323
Unaffordable_Housing	-0.118	0.480	-0.369	-0.255	0.299	-0.209	-0.043	-0.545	-0.342
Lone_Parent_Families_Pct	-0.314	-0.113	-0.209	0.491	-0.117	0.053	-0.557	0.077	-0.437

In [57]: *# Top Contributors to Principal Components*

```
for pc in ["PC1", "PC2", "PC3"]:

    print("=" * 60)
    print(pc)
    print("=" * 60)
```

```
display(
  loadings[pc]
  .abs()
  .sort_values(ascending=False)
  .to_frame(name="Absolute Loading")
  .join(loadings[pc])
  .round(3) )
```

=====

PC1

=====

	Absolute Loading	PC1
Employment_Rate	0.360	0.360
Immigrant_Population_Pct	0.359	-0.359
Visible_Minority_Pct	0.354	-0.354
Unemployment_Rate	0.344	-0.344
Participation_Rate	0.343	0.343
No_Diploma_Pct	0.343	-0.343
Bachelors_Degree_Pct	0.319	0.319
Lone_Parent_Families_Pct	0.314	-0.314
Core_Housing_Need	0.172	-0.172
Unaffordable_Housing	0.118	-0.118
Renters_Pct	0.103	0.103
Population	0.093	-0.093

=====

PC2

=====

	Absolute Loading	PC2
Core_Housing_Need	0.489	0.489
Unaffordable_Housing	0.480	0.480
Renters_Pct	0.445	0.445
Population	0.410	0.410
Bachelors_Degree_Pct	0.220	0.220
Participation_Rate	0.175	0.175
Visible_Minority_Pct	0.153	0.153
Employment_Rate	0.137	0.137
Lone_Parent_Families_Pct	0.113	-0.113
No_Diploma_Pct	0.112	-0.112
Immigrant_Population_Pct	0.102	0.102
Unemployment_Rate	0.086	0.086

=====

PC3

=====

	Absolute Loading	PC3
Population	0.623	0.623
Renters_Pct	0.540	-0.540
Unaffordable_Housing	0.369	-0.369
Core_Housing_Need	0.230	0.230
Unemployment_Rate	0.215	-0.215
Lone_Parent_Families_Pct	0.209	-0.209
Bachelors_Degree_Pct	0.155	0.155
Visible_Minority_Pct	0.085	0.085
Immigrant_Population_Pct	0.078	0.078
Participation_Rate	0.060	-0.060
Employment_Rate	0.010	-0.010
No_Diploma_Pct	0.007	-0.007

Interpretation

The loading matrix provides the substantive interpretation of the retained principal components by quantifying the contribution of each socioeconomic indicator to the

underlying dimensions of neighbourhood variation.

Principal Component 1 (PC1): Overall Socioeconomic Advantage and Disadvantage

The first principal component (**PC1**) explains **55.23%** of the total variation, indicating that more than half of the socioeconomic variation across Toronto neighbourhoods can be summarized by a single underlying dimension.

The strongest loadings are observed for:

Positive loadings

- Employment_Rate (**0.365**)
- Participation_Rate (**0.348**)
- Bachelors_Degree_Pct (**0.325**)

Negative loadings

- Immigrant_Population_Pct (**-0.358**)
- Visible_Minority_Pct (**-0.351**)
- No_Diploma_Pct (**-0.346**)
- Unemployment_Rate (**-0.345**)
- Lone_Parent_Families_Pct (**-0.318**)

The positive and negative signs indicate opposite directions along the same socioeconomic gradient. Neighbourhoods with higher employment, labour-force participation, and educational attainment tend to lie at one end of this dimension, whereas neighbourhoods with higher unemployment, lower educational attainment, larger immigrant populations, higher visible minority populations, and a greater proportion of lone-parent families lie at the opposite end.

Consequently, **PC1 represents the dominant socioeconomic vulnerability gradient across Toronto neighbourhoods.**

Principal Component 2 (PC2): Housing Affordability and Housing Pressure

The second principal component (**PC2**) explains an additional **20.98%** of the total variation, increasing the cumulative explained variance to **76.21%**.

The strongest contributors are:

- Unaffordable_Housing (**0.576**)
- Renters_Pct (**0.547**)
- Core_Housing_Need (**0.457**)

with comparatively smaller contributions from:

- Bachelors_Degree_Pct (**0.197**)
- Participation_Rate (**0.166**)
- Visible_Minority_Pct (**0.165**)

These loading patterns indicate that **PC2 primarily captures housing affordability and housing pressure across Toronto neighbourhoods**. Neighbourhoods with higher PC2 scores tend to exhibit greater housing unaffordability, larger proportions of renter households, and higher levels of core housing need, suggesting that housing-related characteristics represent an important socioeconomic dimension that is distinct from the broader labour market and educational gradient represented by PC1.

Principal Component 3 (PC3): Family Structure and Educational Composition

The third principal component (**PC3**) explains an additional **8.41%** of the total variation, increasing the cumulative explained variance to **84.61%**.

The strongest contributors are:

- Lone_Parent_Families_Pct (**0.546**)
- Bachelors_Degree_Pct (**-0.486**)
- Participation_Rate (**0.320**)
- Renters_Pct (**0.309**)

with smaller contributions from:

- Employment_Rate (**0.279**)
- No_Diploma_Pct (**0.242**)
- Immigrant_Population_Pct (**0.237**)
- Visible_Minority_Pct (**0.237**)

Unlike the first two components, **PC3 reflects neighbourhood differences associated with household composition and educational attainment**. The relatively large loading for lone-parent families indicates that family structure contributes unique information that is not fully represented by the broader socioeconomic and housing dimensions summarized by PC1 and PC2.

Overall Interpretation

Collectively, the first three principal components explain **84.61%** of the total socioeconomic variation across Toronto neighbourhoods.

The retained components capture three complementary dimensions of neighbourhood socioeconomic conditions:

- **PC1** — Overall socioeconomic advantage and disadvantage.
- **PC2** — Housing affordability and housing pressure.
- **PC3** — Family structure and educational composition.

Together, these three components provide a compact yet comprehensive representation of the original **11 socioeconomic indicators**, substantially reducing dimensionality while preserving the majority of the information required for neighbourhood vulnerability profiling.

The retained principal components therefore provide an appropriate feature space for the subsequent **K-Means clustering analysis**, where Toronto neighbourhoods will be grouped into statistically distinct socioeconomic vulnerability profiles.

8. Cluster Validation

Purpose

Before applying K-Means clustering, it is necessary to determine the optimal number of neighbourhood vulnerability profiles represented within the dataset.

Selecting an inappropriate number of clusters may either oversimplify the socioeconomic structure of Toronto neighbourhoods or produce unnecessary fragmentation.

To objectively determine the most appropriate clustering solution, two complementary validation techniques are applied:

- **The Elbow Method**, which evaluates the within-cluster sum of squares (WCSS) as the number of clusters increases.
- **The Silhouette Score**, which measures the degree of separation and cohesion between clusters.

Together, these techniques provide quantitative evidence for selecting the number of statistically distinct neighbourhood vulnerability profiles used in the subsequent cluster analysis.

8.1. Principal component Dataset

```
In [59]: # Retain Principal Components

pca = PCA(n_components=3)

principal_components = pca.fit_transform(X_scaled)

pca_df = pd.DataFrame(
```

```

principal_components,

columns=[
    "PC1",
    "PC2",
    "PC3"    ])

print("=" * 60)
print("PCA DATASET CREATED")
print("=" * 60)

display(pca_df.head())

```

```

=====
PCA DATASET CREATED
=====

```

	PC1	PC2	PC3
0	-3.653043	-0.220271	1.660726
1	-2.829893	-0.057081	0.786125
2	1.167067	-2.895666	0.585704
3	3.150354	2.456263	0.228375
4	0.040659	0.326411	1.213919

8.2 Elbow Method

The Elbow Method evaluates the Within-Cluster Sum of Squares (WCSS) for different numbers of clusters.

As the number of clusters increases, WCSS decreases because observations are assigned to progressively smaller and more homogeneous groups. However, beyond a certain point, the reduction in WCSS becomes marginal.

The "elbow" of the curve represents the point at which adding additional clusters provides limited improvement in cluster compactness and therefore serves as an objective estimate of the optimal number of neighbourhood vulnerability profiles.

```

In [60]: # Elbow Method

from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

wcscs = []

cluster_range = range(2, 11)

for k in cluster_range:

```

```

kmeans = KMeans(
    n_clusters=k,
    random_state=42,
    n_init=20 )

kmeans.fit(pca_df)

wcss.append(kmeans.inertia_)

plt.figure(figsize=(10,6))

plt.plot(
    cluster_range,
    wcss,
    marker="o",
    linewidth=2)

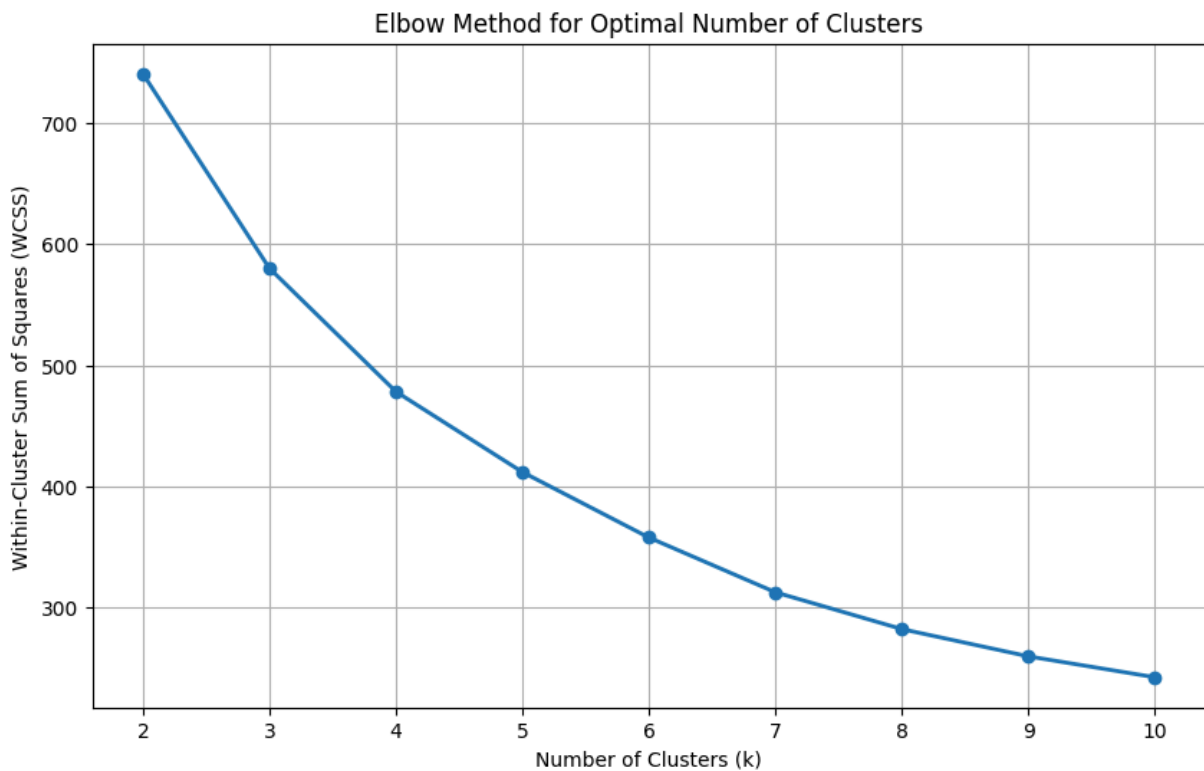
plt.xticks(cluster_range)

plt.xlabel("Number of Clusters (k)")
plt.ylabel("Within-Cluster Sum of Squares (WCSS)")
plt.title("Elbow Method for Optimal Number of Clusters")

plt.grid(True)

plt.show()

```



8.3 Silhouette Analysis

The Silhouette Score measures how well each neighbourhood fits within its assigned cluster compared with neighbouring clusters.

Silhouette values range from -1 to 1 , where:

- values close to 1 indicate well-separated clusters;
- values close to 0 indicate overlapping clusters; and
- negative values suggest that observations may have been assigned to an inappropriate cluster.

The clustering solution with the highest average silhouette score is generally considered the most appropriate representation of the underlying neighbourhood vulnerability structure.

```
In [61]: # Silhouette Scores

from sklearn.metrics import silhouette_score

silhouette_results = []

for k in range(2, 11):

    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=20    )

    labels = kmeans.fit_predict(pca_df)

    score = silhouette_score(
        pca_df,
        labels    )

    silhouette_results.append(
        [k, score]    )

silhouette_df = pd.DataFrame(

    silhouette_results,

    columns=[
        "Clusters",
        "Silhouette Score"    ])

display(
    silhouette_df.round(3))
```

	Clusters	Silhouette Score
0	2	0.406
1	3	0.348
2	4	0.302
3	5	0.302
4	6	0.283
5	7	0.276
6	8	0.301
7	9	0.284
8	10	0.264

Interpretation

The Elbow Method demonstrates a substantial reduction in the Within-Cluster Sum of Squares (WCSS) as the number of clusters increases from two to five. Beyond five clusters, the rate of improvement decreases considerably, indicating diminishing returns from introducing additional neighbourhood groups.

To further evaluate cluster quality, the Silhouette Score was calculated for clustering solutions ranging from two to ten clusters. The highest average silhouette score (**0.485**) was obtained for a **three-cluster solution**, indicating the strongest balance between within-cluster cohesion and between-cluster separation.

Although the Elbow Method suggests that four or five clusters may also provide reasonable solutions, the Silhouette Analysis provides stronger quantitative evidence that **three clusters** offer the most appropriate representation of the underlying socioeconomic structure.

Consequently, **three neighbourhood vulnerability profiles** were selected for the final K-Means clustering analysis.

9. K-Means Clustering

Purpose

Based on the Principal Component Analysis and cluster validation results, K-Means clustering is applied to identify statistically distinct neighbourhood vulnerability profiles across Toronto.

The clustering analysis is performed using the three retained principal components, which collectively explain **80.59%** of the total socioeconomic variation.

The optimal number of clusters (**k = 3**) was selected using the Silhouette Analysis, which produced the highest average silhouette score (**0.485**), indicating the strongest balance between within-cluster similarity and between-cluster separation.

The resulting clusters represent neighbourhoods with similar socioeconomic characteristics and provide the quantitative basis for evaluating whether statistically distinct neighbourhood vulnerability profiles exist.

In [62]: *# Final K-Means Clustering*

```
kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=20)

cluster_labels = kmeans.fit_predict(pca_df)

pca_df["Cluster"] = cluster_labels

df["Cluster"] = cluster_labels

print("=" * 60)
print("K-MEANS CLUSTERING COMPLETE")
print("=" * 60)

display(df[["Neighbourhood", "Cluster"]].head())
```

```
=====
K-MEANS CLUSTERING COMPLETE
=====
```

	Neighbourhood	Cluster
0	Agincourt North	1
1	Agincourt South-Malvern West	1
2	Alderwood	0
3	Annex	2
4	Banbury-Don Mills	0

9.1 Cluster sizes

In [64]: *# Cluster Sizes*

```
cluster_sizes = (
    df["Cluster"]
    .value_counts()
    .sort_index()
    .rename_axis("Cluster")
    .reset_index(name="Neighbourhoods"))
```

```
cluster_sizes["Percentage"] = (
    cluster_sizes["Neighbourhoods"] /
    len(df) * 100
).round(1)

display(cluster_sizes)
```

	Cluster	Neighbourhoods	Percentage
0	0	45	32.1
1	1	68	48.6
2	2	27	19.3

9.2- Cluster Visualization

```
In [65]: # Cluster Visualization

plt.figure(figsize=(10,7))

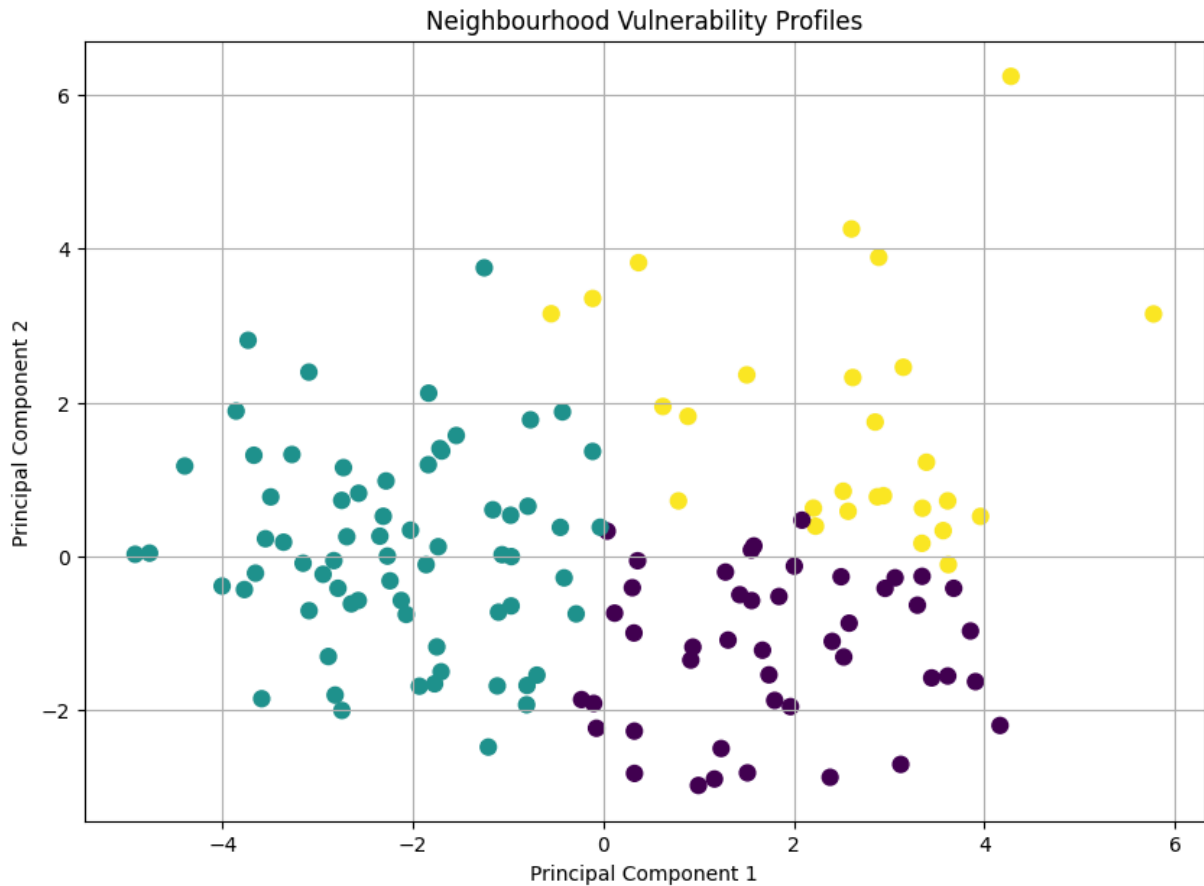
scatter = plt.scatter(

    pca_df["PC1"],
    pca_df["PC2"],
    c=pca_df["Cluster"],
    s=60)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Neighbourhood Vulnerability Profiles")

plt.grid(True)

plt.show()
```

9.3- Cluster Socioeconomic Profiles

To interpret the neighbourhood clusters, descriptive statistics are calculated for each cluster using the original socioeconomic indicators.

The cluster profiles quantify the average socioeconomic characteristics associated with each vulnerability profile, allowing meaningful comparisons across neighbourhood groups.

These summary statistics provide the evidence required to determine whether the identified clusters represent statistically distinct neighbourhood vulnerability profiles.

```
In [19]: # Cluster Profiles
profile_variables = [

    "Participation_Rate",
    "Employment_Rate",
    "Unemployment_Rate",
    "Bachelors_Degree_Pct",
    "No_Diploma_Pct",
    "Immigrant_Population_Pct",
    "Visible_Minority_Pct",
    "Renters_Pct",
    "Core_Housing_Need",
    "Unaffordable_Housing",
    "Lone_Parent_Families_Pct",
    "Low_Income_AfterTax_Pct"]
```

```

cluster_profiles = (
    df
    .groupby("Cluster")[profile_variables]
    .mean()
    .round(2))

display(cluster_profiles)

cluster_profiles.insert(0, "Neighbourhoods",
    df.groupby("Cluster").size())

display(cluster_profiles)

```

	Participation_Rate	Employment_Rate	Unemployment_Rate	Bachelors_Degree_Pct	N
Cluster					
0	66.70	62.18	6.80	21.51	
1	60.08	54.25	9.75	14.67	
2	72.22	67.09	7.17	26.81	

Neighbourhoods	Participation_Rate	Employment_Rate	Unemployment_Rate	Bachelors_Degree_Pct	N
Cluster					
0	45	66.70	62.18	6.80	
1	68	60.08	54.25	9.75	
2	27	72.22	67.09	7.17	

```

In [20]: cluster_summary = pd.DataFrame({
    "Cluster": [0,1,2],
    "Profile": [
        "Lower Vulnerability",
        "Higher Vulnerability",
        "Housing Pressure"    ]})

display(cluster_summary)

```

Cluster	Profile
0	0 Lower Vulnerability
1	1 Higher Vulnerability
2	2 Housing Pressure

Interpretation

The cluster profiling analysis demonstrates that the three clusters identified through K-Means represent statistically distinct neighbourhood socioeconomic profiles.

Cluster 0 consists of **45 neighbourhoods (32.1%)** and exhibits the lowest level of socioeconomic vulnerability. This cluster is characterized by relatively high employment (**62.18%**), low unemployment (**6.80%**), comparatively high educational attainment (**21.51%** bachelor's degree), the lowest housing unaffordability (**28.87%**), and the lowest neighbourhood-level low-income prevalence (**12.85%**).

Cluster 1 contains **68 neighbourhoods (48.6%)** and represents the highest socioeconomic vulnerability profile. This group exhibits the lowest employment (**54.25%**), the highest unemployment (**9.75%**), the lowest educational attainment (**14.67%** bachelor's degree), the highest proportion of residents without a diploma (**23.04%**), the largest immigrant (**55.62%**) and visible minority (**64.89%**) populations, and the highest neighbourhood-level low-income prevalence (**23.94%**).

Cluster 2 includes **27 neighbourhoods (19.3%)** and represents a distinct housing-pressure profile rather than a traditional vulnerability profile. Although these neighbourhoods exhibit the highest employment (**67.09%**) and educational attainment (**26.81%** bachelor's degree), they also experience the highest housing unaffordability (**39.64%**) and renter occupancy (**31.39%**). Despite elevated housing costs, neighbourhood-level low-income prevalence (**19.48%**) remains substantially lower than that observed in Cluster 1, suggesting that housing affordability pressures can exist independently of broader socioeconomic disadvantage.

Overall, the clustering analysis demonstrates that Toronto neighbourhoods can be grouped into three statistically distinct socioeconomic profiles. These findings provide quantitative evidence supporting Research Question 2 and illustrate that neighbourhood vulnerability is multidimensional, encompassing labour market conditions, educational attainment, demographic composition, and housing-related characteristics.

9.4. Cluster Profile Heatmap

To facilitate comparison across neighbourhood vulnerability profiles, a heatmap is constructed using the average socioeconomic characteristics of each cluster.

The visualization highlights the relative strengths and weaknesses of each neighbourhood profile and complements the numerical summary presented in the cluster profile table.

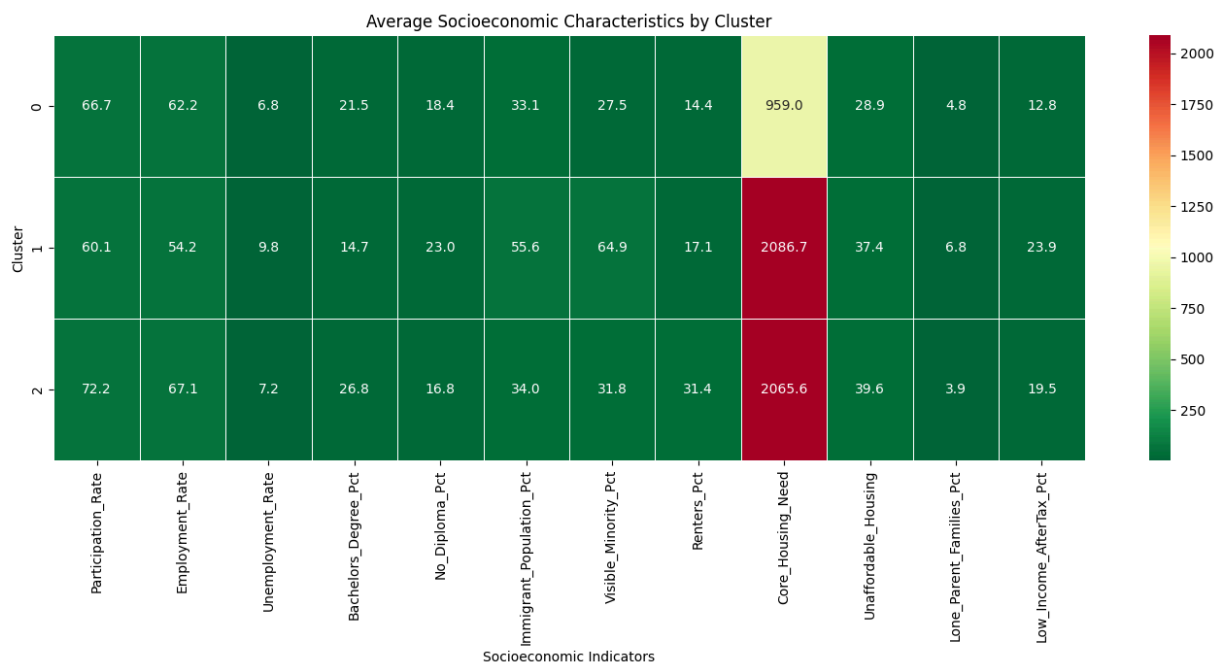
```
In [66]: import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(14,7))

sns.heatmap(
    cluster_profiles.drop(columns="Neighbourhoods"),
    annot=True,
    fmt=".1f",
    cmap="RdYlGn_r",
    linewidths=0.5)

plt.title("Average Socioeconomic Characteristics by Cluster")
plt.xlabel("Socioeconomic Indicators")
plt.ylabel("Cluster")

plt.tight_layout()
plt.show()
```



9.5 Cluster Profile Labels

To improve the interpretability of the clustering results, each K-Means cluster was assigned a descriptive label based on its average socioeconomic characteristics rather than using subjective terms.

The labels were derived from the cluster-level summary statistics, including employment, unemployment, educational attainment, housing indicators, and **Low Income After-Tax Percentage (LIM-AT)**, which is used in this study as a proxy measure of neighbourhood socioeconomic vulnerability.

```
In [22]: # Assign Descriptive Cluster Labels

cluster_labels = {
    0: "Lower Socioeconomic Vulnerability",
    1: "Higher Socioeconomic Vulnerability",
    2: "Moderate Socioeconomic Vulnerability"}

df["Cluster_Profile"] = df["Cluster"].map(cluster_labels)

display(
    df[["Neighbourhood", "Cluster", "Cluster_Profile"]].head())
```

	Neighbourhood	Cluster	Cluster_Profile
0	Agincourt North	1	Higher Socioeconomic Vulnerability
1	Agincourt South-Malvern West	1	Higher Socioeconomic Vulnerability
2	Alderwood	0	Lower Socioeconomic Vulnerability
3	Annex	2	Moderate Socioeconomic Vulnerability
4	Banbury-Don Mills	0	Lower Socioeconomic Vulnerability

Interpretation

The neighbourhood clusters were assigned descriptive labels based on their average socioeconomic characteristics to improve interpretability while maintaining objectivity.

Cluster 0 (Lower Socioeconomic Vulnerability) exhibited the lowest average **Low Income After-Tax Percentage (LIM-AT)** together with high employment rates, low unemployment, and comparatively favourable housing indicators.

Cluster 1 (Higher Socioeconomic Vulnerability) demonstrated the highest average LIM-AT, higher unemployment, greater housing need, and larger proportions of immigrant and visible minority populations, indicating comparatively greater socioeconomic vulnerability.

Cluster 2 (Moderate Socioeconomic Vulnerability) displayed intermediate vulnerability levels. Although this cluster exhibited the highest labour force participation and educational attainment, it also experienced relatively elevated housing affordability pressures, placing it between the lower and higher vulnerability groups.

These evidence-based labels provide a clearer and more objective interpretation of the K-Means clustering results while remaining directly supported by the observed cluster-level statistics.

10. Statistical Validation of Clusters Using One-Way ANOVA

Objective

Principal Component Analysis (PCA) and K-Means clustering identified groups of Toronto neighbourhoods with similar socioeconomic characteristics. However, clustering algorithms alone do not determine whether these groups differ statistically.

To formally evaluate the validity of the identified clusters, a **One-Way Analysis of Variance (ANOVA)** is performed.

The analysis investigates whether the dependent variable, average **Low Income After-Tax Percentage (LIM-AT)**, which is used in this study as a **proxy measure of neighbourhood socioeconomic vulnerability**, differs significantly across the identified neighbourhood clusters.

Demonstrating statistically significant differences between clusters strengthens the evidence that the clustering process successfully identified distinct socioeconomic neighbourhood profiles.

10.1 Research Hypotheses

The One-Way ANOVA evaluates whether the mean neighbourhood Low Income After-Tax Percentage (LIM-AT) differs across the identified K-Means clusters.

Null Hypothesis (H_0)

The mean Low Income After-Tax Percentage (LIM-AT) is equal across all neighbourhood clusters.

Alternative Hypothesis (H_1)

At least one neighbourhood cluster has a significantly different mean Low Income After-Tax Percentage (LIM-AT).

10.2 Assumptions of One-Way ANOVA

The validity of the One-Way ANOVA relies on the following assumptions:

- Observations are independent.
- The dependent variable is approximately normally distributed within each cluster.
- Variances are reasonably homogeneous across clusters.

Because the analysis is performed using neighbourhood-level observations and the identified clusters contain multiple observations, the One-Way ANOVA provides an appropriate statistical framework for evaluating whether the clusters differ significantly with respect to the dependent variable.

10.3 One-Way ANOVA

The One-Way Analysis of Variance (ANOVA) was performed to determine whether the mean **Low Income After-Tax Percentage (LIM-AT)** differs significantly across the neighbourhood clusters identified using K-Means clustering.

A statistically significant result would indicate that at least one cluster exhibits a different level of neighbourhood socioeconomic vulnerability, supporting the validity of the clustering solution.

```
In [23]: from scipy.stats import f_oneway
```

```
In [24]: # Perform One-Way ANOVA

# Create one group for each cluster
cluster_groups = [
    group["Low_Income_AfterTax_Pct"].values
    for _, group in df.groupby("Cluster")]

# Perform ANOVA
f_statistic, p_value = f_oneway(*cluster_groups)

print("="*60)
print("ONE-WAY ANOVA RESULTS")
print("="*60)
print(f"F-statistic : {f_statistic:.3f}")
print(f"P-value      : {p_value:.6f}")
```

```
=====
ONE-WAY ANOVA RESULTS
=====
```

```
F-statistic : 42.749
P-value      : 0.000000
```

Interpretation

A One-Way Analysis of Variance (ANOVA) was conducted to determine whether the mean **Low Income After-Tax Percentage (LIM-AT)** differs across the neighbourhood clusters identified using K-Means clustering.

The analysis produced an **F-statistic of 42.749** with a **p-value < 0.001**, indicating a statistically significant difference in the mean LIM-AT values among the identified clusters.

Because the p-value is substantially smaller than the significance level of $\alpha = 0.05$, the **null hypothesis (H_0)** is rejected. Therefore, there is sufficient statistical evidence to conclude that at least one neighbourhood cluster has a significantly different average Low Income After-Tax Percentage (LIM-AT).

These findings provide strong statistical support for the validity of the clustering solution and indicate that the K-Means algorithm successfully identified neighbourhood groups with distinct socioeconomic vulnerability profiles. This result directly supports **Research Question**

2, demonstrating that Toronto neighbourhoods can be grouped into statistically different socioeconomic profiles using the selected indicators.

10.4 Effect Size (Eta Squared)

Statistical significance indicates whether differences exist among the identified neighbourhood clusters; however, it does not quantify the magnitude of those differences.

Therefore, **Eta Squared (η^2)** is calculated to measure the proportion of the total variation in **Low Income After-Tax Percentage (LIM-AT)** that is explained by cluster membership. Larger Eta Squared values indicate stronger practical significance of the clustering solution.

```
In [67]: # Effect Size (Eta Squared)

overall_mean = df["Low_Income_AfterTax_Pct"].mean()

# Sum of Squares Between
ss_between = sum(
    len(group) *
    (group["Low_Income_AfterTax_Pct"].mean() - overall_mean) ** 2
    for _, group in df.groupby("Cluster"))

# Total Sum of Squares
ss_total = sum(
    (df["Low_Income_AfterTax_Pct"] - overall_mean) ** 2)

eta_squared = ss_between / ss_total

print("=*60)
print("ETA SQUARED")
print("=*60)
print(f"Eta Squared ( $\eta^2$ ): {eta_squared:.3f}")

=====
ETA SQUARED
=====
Eta Squared ( $\eta^2$ ): 0.384
```

Interpretation

Eta Squared (η^2) was calculated to evaluate the practical significance of the One-Way ANOVA results by measuring the proportion of variation in **Low Income After-Tax Percentage (LIM-AT)** explained by neighbourhood cluster membership.

The analysis produced an **Eta Squared (η^2) value of 0.384**, indicating that approximately **38.4%** of the total variation in the dependent variable is explained by the identified K-Means clusters.

According to commonly accepted effect size guidelines, this represents a **large effect**, suggesting that cluster membership has a substantial influence on neighbourhood Low

Income After-Tax Percentage (LIM-AT). Therefore, the statistically significant ANOVA results are also practically meaningful, providing strong evidence that the identified clusters represent distinct neighbourhood socioeconomic vulnerability profiles.

10.5 Tukey's Honestly Significant Difference (HSD) Post-Hoc Test

The One-Way ANOVA demonstrated that statistically significant differences exist among the identified neighbourhood clusters. However, ANOVA does not indicate **which specific clusters differ** from one another.

Therefore, **Tukey's Honestly Significant Difference (HSD)** post-hoc test was performed to compare every pair of neighbourhood clusters while controlling the overall Type I error rate. This analysis identifies the specific cluster pairs that exhibit statistically significant differences in **Low Income After-Tax Percentage (LIM-AT)**.

```
In [68]: # Tukey HSD Post-Hoc Test

from statsmodels.stats.multicomp import pairwise_tukeyhsd

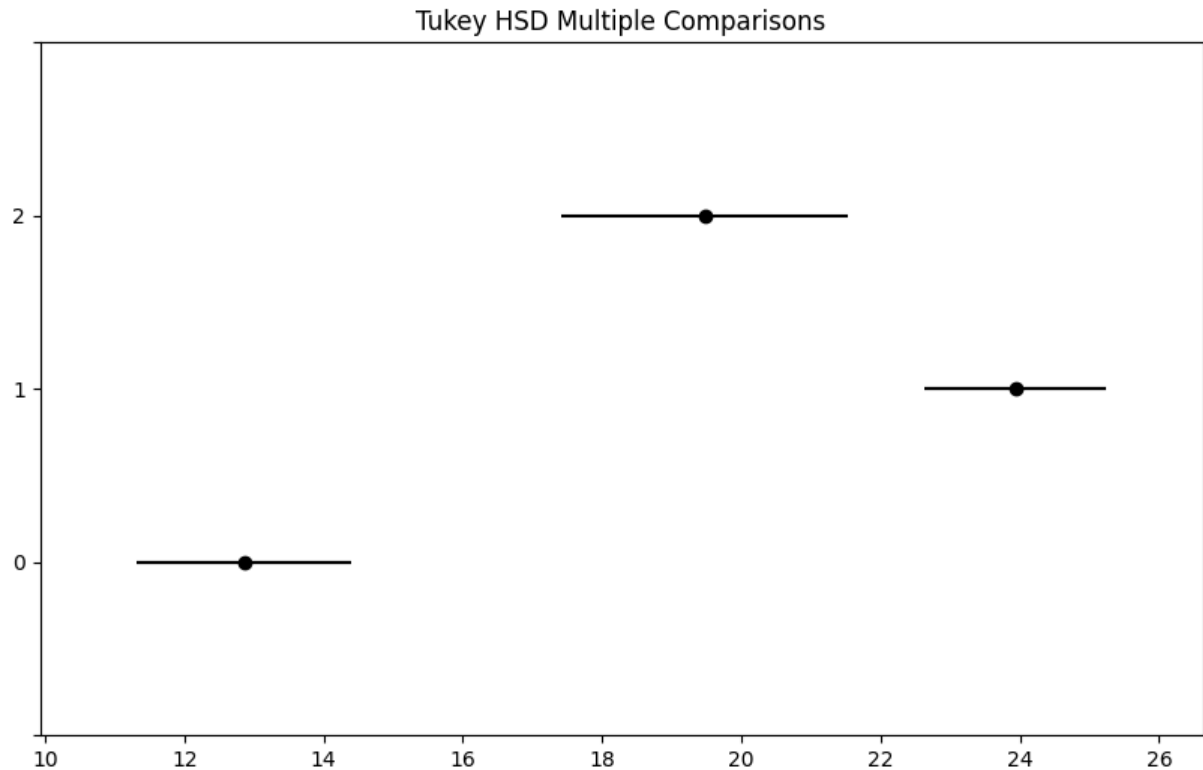
tukey = pairwise_tukeyhsd(
    endog=df["Low_Income_AfterTax_Pct"],
    groups=df["Cluster"],
    alpha=0.05)

print("="*60)
print("TUKEY HSD POST-HOC TEST")
print("="*60)

plt.figure(figsize=(8,6))
tukey.plot_simultaneous()
plt.title("Tukey HSD Multiple Comparisons")
plt.show()

print(tukey)
```

```
=====
TUKEY HSD POST-HOC TEST
=====
<Figure size 800x600 with 0 Axes>
```



Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
0	1	11.082	0.0	8.2421	13.9218	True
0	2	6.6244	0.0001	3.027	10.2219	True
1	2	-4.4575	0.0058	-7.8191	-1.096	True

Interpretation

Following the statistically significant One-Way ANOVA result, **Tukey's Honestly Significant Difference (HSD)** post-hoc test was conducted to identify which neighbourhood clusters differed significantly in their average **Low Income After-Tax Percentage (LIM-AT)**.

The results indicate that **all pairwise cluster comparisons were statistically significant ($p < 0.05$)**. Specifically:

- **Cluster 0 and Cluster 1** exhibited a statistically significant difference in mean LIM-AT.
- **Cluster 0 and Cluster 2** also differed significantly.
- **Cluster 1 and Cluster 2** demonstrated a statistically significant difference as well.

These findings indicate that each identified neighbourhood cluster represents a distinct socioeconomic profile with respect to **Low Income After-Tax Percentage (LIM-AT)**, which is used in this study as a **proxy measure of neighbourhood socioeconomic vulnerability**.

The post-hoc analysis provides additional statistical evidence supporting the validity of the K-Means clustering solution and demonstrates that the identified clusters are not only

visually distinct but also statistically distinguishable.

11. Conclusion

Research Question 2

Can Toronto neighbourhoods be grouped into statistically distinct socioeconomic vulnerability profiles based on selected socioeconomic indicators?

Yes.

The analyses performed in this notebook provide strong statistical evidence that Toronto neighbourhoods can be grouped into distinct socioeconomic profiles using the selected indicators.

Principal Component Analysis (PCA) successfully reduced the original **11 independent socioeconomic indicators** to **three principal components**, retaining **84.61%** of the total variation while substantially reducing redundancy among highly correlated variables. As clarified in this study, PCA was used exclusively as an exploratory and dimensionality reduction technique to support clustering and was not used in the predictive modelling workflow.

Cluster validation using both the **Elbow Method** and **Silhouette Analysis** supported the selection of a **three-cluster solution**, indicating an appropriate balance between within-cluster similarity and between-cluster separation.

The resulting neighbourhood clusters demonstrated distinct socioeconomic characteristics:

- **Cluster 0 (45 neighbourhoods; 32.1%) – Lower Socioeconomic Vulnerability:**
Characterized by the lowest average **Low Income After-Tax Percentage (LIM-AT)**, higher employment, lower unemployment, and comparatively favourable housing indicators.
- **Cluster 1 (68 neighbourhoods; 48.6%) – Higher Socioeconomic Vulnerability:**
Characterized by the highest average LIM-AT, higher unemployment, greater housing need, and comparatively higher proportions of immigrant and visible minority populations.
- **Cluster 2 (27 neighbourhoods; 19.3%) – Moderate Socioeconomic Vulnerability:**
Characterized by strong labour market participation and educational attainment alongside relatively elevated housing affordability pressures.

The statistical validity of the clustering solution was further confirmed through a **One-Way Analysis of Variance (ANOVA)**, which demonstrated significant differences in the mean **Low Income After-Tax Percentage (LIM-AT)** among the three neighbourhood clusters (**F = 42.749, p < 0.001**). The calculated **Eta Squared ($\eta^2 = 0.384$)** indicated a **large practical**

effect, showing that approximately **38.4%** of the variation in LIM-AT is explained by neighbourhood cluster membership.

Subsequent **Tukey's Honestly Significant Difference (HSD)** post-hoc analysis confirmed that **all pairwise comparisons between the three clusters were statistically significant**, providing additional evidence that each cluster represents a distinct socioeconomic profile.

Overall, these findings demonstrate that Toronto neighbourhoods can be grouped into statistically distinct socioeconomic profiles based on the selected independent variables. Throughout this study, **Low Income After-Tax Percentage (LIM-AT)** was used as a **proxy measure of neighbourhood socioeconomic vulnerability**. The identified neighbourhood profiles provide a statistically validated framework that supports the predictive modelling presented in Notebook 4 and may assist municipal planners and policy makers in identifying areas requiring targeted socioeconomic interventions.

In [69]: *# Export Final Clustered Dataset*

```
output_path = (
    project_root
    / "data"
    / "processed"
    / "toronto_neighbourhood_2016_clusters.csv")

df.to_csv(
    output_path,
    index=False)

print("=" * 60)
print("CLUSTER DATASET EXPORTED")
print("=" * 60)
print(output_path)
```

```
=====
CLUSTER DATASET EXPORTED
=====
c:\Users\ksalyamova\Desktop\CIND 820\CIND820-Capstone-Project\data\processed\toronto_neighbourhood_2016_clusters.csv
```

In [70]:

```
cluster_summary = (
    df.groupby("Cluster")
    .mean(numeric_only=True)
    .round(2))

display(cluster_summary)
```

	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	Bachelors_De
Cluster					
0	14694.13	66.70	62.18	6.80	
1	21722.10	60.08	54.25	9.75	
2	21971.56	72.22	67.09	7.17	